# Chapter 1

# Protein–Ligand Interactions: Fundamentals

**Mark A. Williams**

## Abstract

Here are described the basic mechanisms governing the interactions between proteins and their natural or manmade ligands, together with the principles underlying their analysis. The consequences of these principles are detailed for the simplest case of one-to-one binding. The general features of experimental measurements of biomolecular interactions arise from properties of the molecules involved and, thus, are common to many methods of detection. Consequently, an understanding of these principles greatly simplifies adoption and comparison of experimental methods and provides the rationale underlying many common protocols. In seeking to understand and interpret the results of experiments or identify possible sources of error these fundamental ideas are a constant guide.

**Key words** Protein complexes, Thermodynamics, Kinetics, Nonlinear regression, Specificity, Drug design

## 1 Introduction

### *1.1 Protein Interactions Are Key to Biological Function*

Proteins have many functions: as enzymes that accelerate vital chemical reactions, as regulatory molecules whose binding to other proteins inhibits or activates their function, as components of the microscopic structures of the cytoskeleton, as engines whose conformational changes act as molecular motors, as signals via their own chemical modification, as detectors of signals through binding of small molecules, peptides or proteins, as selective channels that enable transport through cellular membranes. In all of these roles, proteins bind small molecules, nucleic acids, and other protein partners forming a transient or long-lived complex through noncovalent interactions between the components of the complex. These interactions are often very specific in that a particular protein binds only to one or a few other molecules in the cell. Many cellular processes are performed by a group of proteins which act in a coordinated manner, for example, in metabolic pathways in which an initial molecule is modified by several enzymes in a series of steps, or in a signaling pathway in which binding of an extracellular

Mark A. Williams and Tina Daviter (eds.), *Protein-Ligand Interactions: Methods and Applications*, Methods in Molecular Biology, vol. 1008, DOI 10.1007/978-1-62703-398-5_1, © Springer Science+Business Media New York 2013

signal causes a conformational change which enables a kinase to become active and phosphorylate a target (or targets), which then leads to their binding other proteins and so on. Such networks of interactions between proteins are closely regulated. Changes in the populations of particular protein complexes, or changes in populations of the products of protein mediated reactions, of as little as a factor of two can switch cells from one state to another; from growth to stasis, from replication to apoptosis.

Such small variations in molecular populations correspond to very small differences in the thermodynamics or kinetics of reactions. Consequently, detailed characterization of protein interactions with their binding partners is of paramount importance in a quantitative and integrative biology which aims to understand biological systems in terms of their molecular components. Perhaps the ultimate goal is to characterize the thermodynamic and kinetic behavior of components of living systems in sufficient detail that the response of those systems to stimuli (be they natural substrates or drugs) can be modeled with sufficient accuracy that the behavior of the system may be predicted or modified as desired.

### 1.2 Important Characteristics of Protein–Ligand Interactions

Key to the function of individual proteins and the network of interactions to which they belong are three factors (1) how much protein and ligand are present, (2) how much complex is formed, and (3) how quickly do the complexes form and break apart. Knowing these facts for each protein and complex enables, in principle, the prediction of behaviors of whole networks of interactions. It also enables the prediction of the outcome of any modification in the behavior of one reaction (e.g., as a result of mutation or addition of a competitive ligand which changes the amount of a complex formed) on the behavior of the system as a whole (1, 2). Biophysical approaches to protein–ligand interactions are concerned with identifying the molecular components of complexes (*see* Fig. 1) and quantifying their equilibrium populations and kinetics (rates) of association and dissociation. The need for information on these characteristics is common to all types of interaction (i.e., between any pair of—protein, small molecule, and nucleic acid). Enzymes are a special case, which, in addition to being characterized by each of their binding reactions with substrates, products, cofactors, activators, and inhibitors, are characterized by the degree of acceleration of each reaction that they enable. Only their binding reactions will be considered here.

Let us consider the simple binding reaction

$$\mathrm{P} + \mathrm{L} \rightleftarrows \mathrm{PL}, \tag{1}$$

in which each protein, P, binds one ligand, L, to form a complex, PL, in a reversible manner.

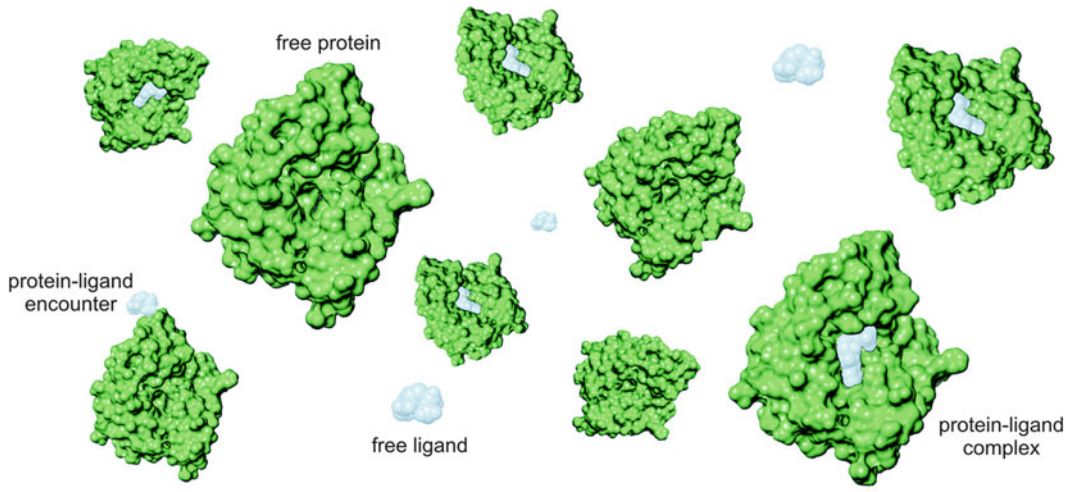


**Fig. 1** Molecular-scale view of a binding equilibrium. In all circumstances there is a mixture of each of the molecular species, free (unbound) protein, free ligand, and the protein–ligand complex that becomes populated as the ratio of ligand to protein increases. The protein–ligand complex will comprise ligand bound at a particular site on the protein's surface. Other "encounter" complexes of distinct conformation will form transiently, but no one of these conformations will have a significant population

At equilibrium, by definition, there are no changes in the concentration of any components of the solution, which means that the rate of association of free protein and free ligand molecules to form complexes is equal to the rate of dissociation of complexes into their constituent molecules.

The overall rate of association will depend on the concentration of both the free ligand, [L], and free protein, [P], present in solution, because the more ligand and protein molecules there are, the more likely they are to come close to (encounter) each other and associate. A process involving collision of two molecules is described by "second order" kinetics, the second order association rate constant, $k_{on,}$ in units of $M^{-1}$ $s^{-1}$, describes the rate of association for a (hypothetical) standard state of 1M protein and 1M ligand. In an idealized, purely diffusion controlled reaction, the on-rate constant depends only upon the size of the molecules involved (their translational and rotational diffusion coefficients) and the size of the binding site; $k_{on,}$ is ~$10^8$ $M^{-1}$ $s^{-1}$ for a medium size protein and a small-molecule ligand (3). The initial encounter rate will be greater for larger proteins, because of their greater surface area, perhaps by a factor of 10–100 (4, 5). For a particular real protein–ligand interaction, $k_{on,}$ is an intrinsic property of the individual molecules involved, which depends not only upon their size but also the microscopic details of their interaction. Not all close encounters are productive in terms of forming the final complex, but it seems that many proteins have evolved to enhance the chance of their cognate ligand binding (i.e., to increase $k_{on}$) once

protein and ligand are near to each other. If a charged protein and ligand approach through random diffusion to within a few nanometres, long-range electrostatic interactions can then pull them toward each other and steer the charged ligand toward the binding site (3, 6, 7). Alternatively, the arrangement of surface chemical groups on the protein may guide a ligand that collides anywhere on the protein surface toward the binding site (8). In contrast, $k_{on}$ may be reduced in a situation where only a subset of the protein or ligand molecules are in a binding competent conformation at any one time or which have unfavorable electrostatics. Such enhancement or reduction of $k_{on}$ may serve a functional purpose in native interactions or may be manipulated by the experimenter through chemical modification or mutation to achieve some desired outcome (9, 10).

The dissociation rate constant, $k_{off}$, is simply a rate per second, and does not depend on concentration because it represents the probability with which each individual complex that is present in solution dissociates in the next second (a process dependent on only a single species is called first order). Dissociation is usually stochastic and depends upon the ligand having excess local energy (gained through random "collisions" with surrounding protein atoms or solvent) sufficient to escape from the protein's binding site. Although the actual lifetime of any particular complex will vary randomly, the mean lifetime $\tau = 1/k_{off}$ averaged over many complexes will be well-defined and can be used to describe behavior of populations of molecules. $k_{off}$ is strongly dependent on the nature of interactions in the bound state. It is presumed that where there are a few individual strong interactions between chemical groups of the ligand and protein, they create a single substantial activation barrier to release, which is rarely crossed because it is rare that a sufficiently large amount of energy will be available, and consequently $k_{off}$ is small. Conversely, in a case where there are many weak interactions contributing to binding, these can each be broken successively with relatively small amounts of energy and $k_{off}$ is relatively large.

The rates of the association and dissociation reactions observed in a particular experiment are equal to the rate constants multiplied by the appropriate concentrations of each species. At equilibrium there is no change in the system and the rate of association equals the rate of dissociation, i.e.,

$$k_{on}[L][P] = k_{off}[PL]. \tag{2}$$

At equilibrium, the ratio of concentrations of the free protein and ligand molecules to the bound complex defines the association constant ($K_a$ or binding affinity). Rearranging Eq. 2 gives

$$K_a = \frac{k_{on}}{k_{off}} = \frac{[PL]}{[P][L]}, \tag{3}$$

where the units of $K_a$ are $M^{-1}$. Equivalently, the dissociation constant $K_d$ is also used

$$K_d = \frac{k_{off}}{k_{on}} = \frac{[P][L]}{[PL]}. \quad (4)$$

The units of $K_d$ are Molar, and are frequently in the range of μM to nM, with the latter representing a higher affinity, less dissociable interaction. For a protein–ligand interaction with a diffusion-limited association rate constant of $10^8$ $M^{-1}$ $s^{-1}$ and a typical dissociation constant of 1 μM this implies a $k_{off}$ of 100 $s^{-1}$. On-rate constants for protein–ligand interactions range from $10^4$ $M^{-1}$ $s^{-1}$ to $10^{10}$ $M^{-1}$ $s^{-1}$ and $k_{off}$ from $10^{-4}$ $s^{-1}$ to $10^5$ $s^{-1}$ (11, 12).

### 1.3 Drugs Act by Modifying a Protein's Biological Interactions

Almost all pharmaceuticals act by binding to particular proteins, their "targets", interfering with the protein's biological function. Where the target is an enzyme, drugs are usually inhibitors (antagonists) targeted to bind in such a way that they, at least partially, block the binding-site for the natural substrate, thus inhibiting the enzyme from performing its function. Occasionally, a drug may act in an allosteric manner; binding elsewhere on the protein, but in such a manner as to disturb the (average) conformation of the substrate site so modifying its affinity for the substrate. With other types of protein target, many drugs are agonists; mimics of the natural biological ligand that binds and activates the protein.

### 1.4 Thermodynamics of Protein–Ligand Interactions

The experimentally determined equilibrium and rate constants for protein–ligand interactions can be used to infer some features of the energetics of the interaction between protein and ligand on the microscopic scale. The equilibrium dissociation constant determined under normal experimental conditions of constant temperature and pressure is related to the difference between the free energy, $G$, of protein and ligand molecules alone in solution and when bound together i.e., the binding free energy change $\Delta G_{bind}$ given by

$$\Delta G_{bind} = G^{PL}_{solution} - (G^{P}_{solution} + G^{L}_{solution}). \quad (5)$$

The relationship between $\Delta G_{bind}$ and $K_d$ is usually given (*see* **Note 1**) as

$$\Delta G_{bind} = RT \cdot \ln(K_d), \quad (6)$$

where $R$ is the universal gas constant and $T$ the temperature (*see* **Note 2**). Negative free energy changes favor the reaction; a $K_d = 1$ μM corresponds to $\Delta G_{bind} = -34.2$ kJ $mol^{-1}$. In principle, this free energy change can be related to the redistribution of energy among the atoms of the protein and ligand molecules, and the solvent which surrounds them, upon formation of the complex.

$$\Delta G_{\text{bind}} = RT \cdot \ln\left(\frac{\sum\limits_{\text{freestates}} e^{-\varepsilon_i/RT}}{\sum\limits_{\text{boundstates}} e^{-\varepsilon_i/RT}}\right), \tag{7}$$

where $\varepsilon_i$ are the possible microscopic energy states of the protein, ligand, and the surrounding solvent. Protein, ligand, and water (and other solvent) molecules are in constant motion on the Ångstrom length scale and although the "bound state" may present as single experimental observable (e.g., in respect of properties such as its UV absorption or fluorescence spectrum or NMR frequencies), it is in fact an average over the many bound molecules in a typical sample (i.e., 1 ml of 10 μM protein contains ~ $6 \times 10^{15}$ protein molecules). Both the bound and the free states of the molecules are composed of many distinct possible conformations of protein and ligand and surrounding solvent molecules, each of which will have a particular energy that will contribute to the observed binding free energy. This microscopic heterogeneity of bound and free states is one of the factors that make prediction of thermodynamic and kinetic properties of protein–ligand interactions from the structures of molecules a still unsolved problem (*see* Subheading 1.8 for further discussion).

The free energy change itself can be decomposed into two terms, the enthalpy change, $\Delta H$, and the entropy change, $\Delta S$, of the reaction

$$\Delta G_{\text{bind}} = \Delta H_{\text{bind}} - T\Delta S_{\text{bind}}. \tag{8}$$

Rearrangements of the molecules upon binding lead to changes in the energy states of the system of molecules, and heat is either taken up or released into the surroundings. In the microscopic view (*see* **Note 3**) the molar enthalpy change is

$$\Delta H_{\text{bind}} \approx \Delta E_{\text{bind}} = \frac{\sum\limits_{\text{boundstates}} \varepsilon_i \cdot e^{-\varepsilon_i/RT}}{\sum\limits_{\text{boundstates}} e^{-\varepsilon_i/RT}} - \frac{\sum\limits_{\text{freestates}} \varepsilon_i \cdot e^{-\varepsilon_i/RT}}{\sum\limits_{\text{freestates}} e^{-\varepsilon_i/RT}}. \tag{9}$$

If the surroundings of the reaction volume consist of a microcalorimeter, then the heat transfer can be measured and used to monitor the reaction and the total heat lost or gained per mole of complex formed is $\Delta H_{\text{bind}}$ (*see* Chapter 4 for further details). It is important to remember that the enthalpic and entropic changes separately do not determine the outcome of the reaction, only the free energy change corresponds to the changes in population of bound and free species. For any given reaction, either or both of $\Delta H_{\text{bind}}$ or $-T\Delta S_{\text{bind}}$ may be a favorable (i.e., have negative value) or unfavorable contribution. However, because, by definition, reported interactions have a negative $\Delta G_{\text{bind}}$ it is most common that both $\Delta H_{\text{bind}}$ and $-T\Delta S_{\text{bind}}$ are favorable.

The $\Delta H_{bind}$ and $-T\Delta S_{bind}$ values are additional characteristics of a reaction that may in principle provide some mechanistic insight. For example, because the measured heat transfer associated with binding arises from all the events that occur in the reaction vessel and because changes in protonation state cause heat changes, it is straight forward to detect protonation state changes that accompany binding using calorimetry. The reaction is simply carried out several times at different pH values or in buffers with different heats of ionization and an observed change in the heat of reaction implies that a protonation change occurs. Less straightforwardly, there are good theoretical reasons to suppose that precisely complimentary interactions between protein and ligand will generate large favorable enthalpy changes upon binding. Thus, modifications of a ligand that make $\Delta H_{bind}$ more negative may indicate improved complimentary, which would likely also mean greater specificity of interaction (*see* Subheading 1.10), which is a desirable property in drugs (13). Practical application of this scheme in drug development is, however, complicated by the role of water molecules, which could have large positive or negative enthalpy contributions in any particular protein–ligand reaction, and by the phenomenon of enthalpy-entropy compensation.

Experimental evidence shows that changes in $\Delta H_{bind}$ and $-T\Delta S_{bind}$ due to modification of an interaction (e.g., through changes to ligand chemistry or amino-acid mutation) tend to be compensated (i.e., change in opposite directions) for most protein–ligand interactions, leaving the $\Delta G_{bind}$ only slightly changed (14). Enthalpy-entropy compensation and the poorly understood role of water are significant (and possibly related) barriers to straight-forwardly interpreting or utilizing the components of the free energy in an explanatory manner at the present time (this does not prevent many publications purporting to do this!). The origin of enthalpy-entropy compensation is not precisely clear, but must lie in the limited number of ways in which energy can be distributed in a particular protein binding site (15). Consequently, the major present use of calorimetry is simply as a method to obtain affinities and the $\Delta H_{bind}$ and $-T\Delta S_{bind}$ data are largely being accumulated as a challenge to computational modeling and for their potential to provide insight in the future.

### 1.5 Typical Properties of Natural Ligands and Drugs

Most biological interactions are required to be functionally reversible in vivo. For example, where a protein binds to an activating phosphopeptide and a signal is propagated (e.g., by conformational change and binding to another protein), phosphatase activity may then increased via some cellular feedback circuit, and the activating phosphopeptide must then escape from its binding site in order to be dephosphorylated so that signal can be switched off. In the case of enzymes, substrate must bind rapidly and product must be released fairly soon after its formation. Further, many enzyme

cofactors (ATP, NADH, FADH) require regeneration by another enzyme and must be released promptly to solution to enable regeneration. Consequently, most proteins are evolved so that their natural substrates are able to bind and be released, and rebind reasonably frequently at cellular concentrations. This means, for example, that $k_{off}$ should not be much slower than 1 $s^{-1}$ if the system is required to respond to changes on the timescale of seconds or not slower than 1,000 $s^{-1}$ if it operates on a millisecond timescale. It is also necessary that regulated variation in protein concentration (e.g., through control of gene expression or of protein degradation) substantially and proportionately modulates the population of complexes present. As will be seen below in Subheading 2.1, this means that the $K_d$ of a protein–ligand interaction needs to be similar to the cellular concentration of that protein. Experimentally it is found that almost all protein interactions with their natural substrates have a $K_d$ between 50 nM and 1 mM (16), commensurate with cellular protein concentrations which are in the range 5 nM to 100 μM in the vast majority of cases (17).

Any drug whose mode of action is to prevent binding of the natural substrate, must replace more than 90 % of substrate. If substrate and drug are present at the same concentration, then the drug must have a tenfold greater affinity than substrate. If the substrate concentration is tenfold greater than that of the drug, the drug must have a 100-fold greater affinity. Of course, if substrate concentrations are relatively low or the drug is designed to supplement natural substrate activity, then its affinity may in principle be more substrate-like. However, often there are also advantages to lowering drug concentration as this may reduce side-effects. Creating high-affinity ligands is thus an explicit goal in the pharmaceutical industry. Consequently, drugs typically have greater affinity for their protein targets than natural substrates, most frequently $K_d$ is in the 100 nM to 10 pM range. Drugs may also be developed to have a slower off rate constant, thus prolonging activation. Furthermore, whereas most interactions with natural ligands have $\Delta H_{bind}$ as the largest favorable contribution, drugs on average have a larger favorable $-T\Delta S_{bind}$ term and binding is typically driven by both favorable enthalpy and entropy of binding—this is likely a consequence of drugs being on average more apolar than natural ligands, and bias in the drug design process toward apolar compounds in order to generate affinity or to improve absorption through the cell membrane (16).

### 1.6 Experimental Monitoring of Protein–Ligand Interactions

In principle, any signal that changes in proportion to the amount of complex formed can be used to monitor binding and determine thermodynamic and kinetic characteristics of the reaction. Consequently, there are many techniques in use to characterize protein

binding and new methods emerge every few years. The methodology chosen to investigate a particular protein system will depend upon factors such as the quantities of protein and ligands available, the specific properties of the protein and any known ligands (e.g., are they fluorescent?), and whether precise quantitation is required or simply a yes/no answer to the question "Do they form a complex?" Key to any useful experimental method is an ability to detect signal variation upon complex formation using only small quantities of protein and ligand. Many experimental strategies rely on changes in spectral properties of protein or ligand e.g., UV or visible light absorption, circular dichroism, or fluorescence spectra. Such approaches are naturally limited to those protein–ligand systems in which there are substantial changes in such spectral properties and that is by no means all systems. In contrast, nuclear magnetic resonance (NMR) spectra are extremely sensitive to interactions between chemical groups and can be relied upon to change when a complex is formed. Consequently, NMR is very widely used in routine screening applications as well as investigation of detailed structural features of complexes. The main drawback to NMR is the relatively large quantities of protein required for experiments (usually several milligrams). Because approximately 95 % of protein–ligand reactions will generate sufficient heat change as to be reliably detected by modern calorimeters, isothermal titration calorimetry is also nearly universally applicable, and requires somewhat less protein than NMR. A wide range of methods have also been developed that do not rely upon a change in the properties of the components of the complexes, but instead immobilize one of the putative binding partners and detect the change in location of the other as it binds. Examples of these methods are surface plasmon resonance and optical wave guide technologies. These location-based methods also have near universality and typically require very small quantities of protein, albeit they rely on being able to tether one component to a surface without interfering with the binding event of interest. The applicability of particular experimental methods and common experimental scenarios are considered in more detail in Chapter 3 and specific details of individual methods of detection are given many other chapters.

Although experiments may vary greatly in their mechanism of detection, the main factors in the design of experimental protocols and evaluation of data have many characteristics in common because they depend upon general features of binding reactions and not the instrumentation. Experimental strategies can be developed for detection of formation of a complex via monitoring a change in a ligand or protein property on binding, or through detection and quantitation of the populations of free ligand or free protein. Obviously, analyses of these three types of signal are all simply related by Eq. 3 and can in principle be converted from

one to the other provided total ligand and total protein concentration are known. However, in a typical experimental scenario in which ligand is titrated into protein, for a high-affinity interaction almost all titrated ligand will bind to protein during the initial points of a titration and free ligand concentration will only become substantial as the protein binding sites become saturated late in the titration. In such a scenario, errors are proportionately smaller for methods which directly detect the amount of complex formed or the reduction in free protein. Consequently, only analysis of these types of measurement will be considered in detail in Subheading 2.

### 1.7 Structural Basis of Protein–Ligand Interactions

Proteins usually bind their biological ligands at a single site (rarely more) on a structured domain. Where a biological function requires a protein to bind several distinct ligands, the protein will likely consist of several distinct domains each of which binds a specific "cognate" ligand. There are some notable exceptions to this "rule", such as those enzymes that have cofactor and substrate binding at neighboring sites on a single domain (although it could be argued that these neighbors form a single extended binding site). Ligand-binding sites are often only a very small part of a protein's surface (*see* Fig. 2a) and modification by mutation may change the binding site and affect the affinity or specificity for ligands with little impact on the structure of the domain as a whole. The possibility of combining distinct binding domains in a new protein through gene fusion and the mutability of binding sites means that evolutionarily related proteins can bind distinct ligands generating beneficial new functions (18, 19). A fundamental problem of both molecular biology and drug design is to understand the physical origin of the affinity and specificity of the interactions made by protein domains in terms of interaction of chemical groups of the protein, of the ligand and with surrounding water molecules, and thus to fully understand both the ability of proteins to discriminate between ligands and the physical factors limiting evolution.

In structures of protein complexes, complementarity of the shape (steric complementarity) and of the chemical nature of the groups on the ligand and protein surfaces is generally observed. Thus, complementarity appears to be a key ingredient in the formation of stable complexes and underlies many computational efforts to predict which ligands will bind with high affinity to a given protein. Apolar groups tend to be brought together with apolar groups, hydrogen bond donors match acceptors and charge groups on a ligand are frequently neutralized by a nearby, oppositely charged protein sidechain (*see* Fig. 2b). However, it should always be borne in mind that the precise thermodynamics and kinetics of association are determined not only by the changes in protein–ligand interactions as the protein and ligand come together but also by the changes in protein–water and

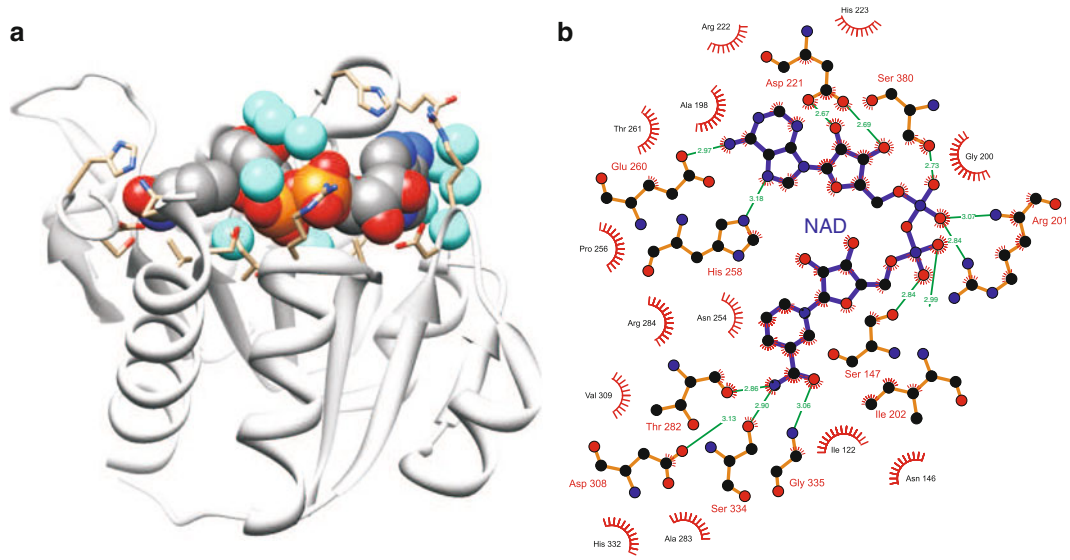


**Fig. 2** The structural basis of ligand recognition by proteins is illustrated here by NAD binding to formate dehydrogenase (PDB structure: 2NAD). (**a**) The path followed by the protein backbone is represented by the ribbon, highlighting the Rossmann-fold architecture common to many nucleotide binding proteins. At the periphery of the protein, conserved amino acids (stick representation) far apart in sequence are found close in space and envelope the dinucleotide (*spheres*), making more than 40 interactions directly or indirectly via crystallographically observed water molecules (*cyan spheres*). (**b**) The direct interactions are annotated in a schematic of the binding site (adapted from PDBsum). Amino acids making hydrogen bonds (*green*) to the ligand are drawn as atoms + bonds, and those making only van der Waals interactions are represented by "eye-lash" symbols (*see* Table 1 for references)

ligand–water interactions. Consequently, it is the net effect of changes in interactions and not complementarity alone which determines ligand binding (20).

It has been inferred from analysis of structures of the free and bound states of protein and ligand, together with experimental thermodynamic data on formation of the complex, that in most cases the largest contribution to affinity is provided by bringing apolar groups (usually $CH_2$, $CH_3$, aromatic, and other carbon-ring structures) of protein and ligand into close proximity (16). The interactions between protein and ligand apolar groups (largely van der Waals interactions) may themselves be energetically more favorable than those they make with water in the free state, but the association is thought to be largely driven by the favorable free energy contribution from water molecules being displaced from the apolar surface into the bulk water where they can form additional hydrogen bonds with other water molecules. It has often been assumed that the driving force of this contribution to the affinity is the gain of entropy due to water release. However, the experimental support for this idea is not strong and it is probably an

oversimplification (14). In practice, there is typically a fine balance between the enthalpic and entropic contributions of the changes in interactions of many individual chemical groups and solvent molecules to overall affinity (16).

The formation of hydrogen bonds between protein and ligand appears to be crucial to the specificity of ligand recognition by proteins. Proteins have many medium-strength hydrogen bond donor and acceptor groups (e.g., N−H, C=O, O−H, COO−, $NH_3$+) and also groups that are able to form weaker bonds (e.g., S−H, C−H, delocalized electrons of aromatic rings) (21). In the free state, almost all of the protein's medium strength hydrogen bond donor and acceptor groups will either make intra-protein hydrogen bonds or hydrogen bond to water molecules (22, 23). Mutational analysis of protein–ligand complexes shows that displacement of water from an acceptor or donor group as a result of binding, without the formation of a new hydrogen bond in the complex, produces a substantial (~5 kJ $mol^{-1}$) unfavorable contribution (24). Hydrogen bonds are also directional (unlike apolar interactions which merely require that atoms are brought close together) and chemical groups from ligand and protein must be oriented appropriately in order to make the bond. Because donor and acceptor groups are generally a minor component of the ligand, there will be only a few relative orientations of protein and ligand which can maximize hydrogen bonding. This directionality and the energetic penalty of not forming hydrogen bonds means that they are key discriminators between ligands and a considerable source of specificity in interactions (25). However, the requirement for complementarity of hydrogen bonding between protein and ligand is not absolute, as interactions between protein and ligand can also be bridged by water molecules, e.g., a ligand NH may donate to water which may then either donate or accept from a polar group on the protein.

### 1.8 Computational Approaches to Prediction of Protein–Ligand Interactions

Although, much has been learnt about the structural features of protein–ligand interactions over the past 30 years, this knowledge has yet to be used with full effect in the interpretation and control of protein function. Statistical analyses of the structures of proteins, protein–ligand complexes, and organic chemicals more widely, have led to a good understanding of the intra-molecular geometries of molecules and also of the intermolecular spatial distribution of interacting chemical groups. Such analyses are an important input to software used to predict the structures of protein complexes and aid in designing chemical modifications of ligands. Molecular modeling tools for "docking" prospective ligands into known active sites and screening for ligands that may be able to form complimentary interactions are now widely and successfully used in ligand discovery and drug design (*see* Chapters 18 and 19).

Presently, it is the case that quantitative analysis of many systems has led to the ability to frequently predict realistic poses (conformations) for small molecules bound to proteins (26) using one of several distinct empirical scoring schemes that partition interacting groups and molecular properties into broad categories (e.g., polar/apolar). These empirical schemes are parameterised to estimate thermodynamic quantities as a weighted sum of changes in types of interaction between the bound and the free state, and are somewhat successful at predicting binding affinities of ligands (27). However, they remain subject to a substantial margin of error in $\Delta G_{\text{bind}}$ (~15 kJ mol$^{-1}$), do not, or poorly, quantitate the enthalpic or entropic components and take no or little account of environmental effects. Consequently, a major remaining challenge is to accurately relate the three-dimensional structures of proteins and ligands to the kinetics and thermodynamics of their interaction.

Future insight will probably arise from developments in more computationally intensive models involving physics-based simulation of atomic interactions and motions and direct averaging over the microscopic states of the protein–ligand system. Such simulations also have the advantage of providing information on the flexibility of the protein target and alternative conformations of the binding site that may not be apparent from a crystal structure. These physics-based methods have been a very active and complex area of research (28) for three decades, but they have not achieved a clear advantage over the cruder empirical methods in predicting affinity or affinity changes. A predictive accuracy of better than ±1 kJ mol$^{-1}$ would be needed to replace experimental study and this still seems some way off.

### 1.9 The Extended Interface: Long-Ranged and Environmental Effects on Ligand Binding

Protein complex formation is usually monitored in vitro in particular buffer and salt solutions that have been found to keep the protein and/or ligand soluble, and in vivo proteins are found in the distinct dense soups of the cellular cytoplasm or blood plasma. Many of the interactions between protein, ligand, and water are sensitive to differences between these environments. The measured affinity and kinetics of protein–ligand interactions depend upon the ionic environment, on pH and on crowding effects due to surrounding nonparticipant proteins or other solvent components. Changes in $K_d$ by a factor of 10–100 in response to a change of 1 pH unit or of ±100 mM NaCl (cellular concentrations are often ~200 mM, but very dependent on the organism) are commonplace (29). Substantial differences in binding affinity between mutant proteins or between different ligands may sometimes be due to their effect on increasing or diminishing the sensitivity to some environmental factor and thus will not correspond to a readily identifiable structural feature of the protein–ligand interaction. Environmental effects are, generally, rather poorly understood and the field as a whole would benefit from more data.

One practical consequence of this sensitivity to environmental factors is that assays of protein–ligand binding must be performed under precisely controlled conditions so that variations in behavior can be reliably attributed to changes in the protein or ligands themselves. A significant feature in the binding of many charged ligands (including common cofactors such as ATP) is change in the protonation state of the ligand or protein on binding. For example, if there is a possibility to form a hydrogen bond in the complex that is stronger than that made with water, then the proton involved in that hydrogen bond will be stabilized in the complex. In a hypothetical case where the hydrogen bonding group has p*K*a = 6 in the free state and in the complex p*K*a = 8, then at any pH between these values, protons will migrate from the solvent to form the hydrogen bonds in the complexes. In general, protonation effects will have an effect both on the free energy and enthalpy of binding, whose pH dependence will reveal the existence and details of any such process for a protein–ligand system of interest. Such mechanistic understanding of environmental influences may lead to more reliable data, identification of better assay conditions (e.g., further from the relevant p*K*as or with higher buffer concentrations) or insight that may guide further modification of the ligand in a drug development program (e.g., to change its charge or p*K*as).

### 1.10 Specificity

It is common to encounter discussions of specificity of interactions (sometimes called selectivity), which encompass several distinct but related ideas concerning the relative likelihood of observing particular complexes. Firstly, the question "Is there a "specific" binding site on the protein for a particular ligand?" means is there a particular location at which almost all the molecules of particular ligand bind during a titration before a population of protein complex is found with the ligand at another location. Kinases specifically bind ATP as a cofactor at an evolutionarily well-conserved site. Secondly, the question may be "Is the protein specific for a particular ligand?" i.e., are there no other small molecules (in the cell) which will bind at the same site. In the case of kinases other adenine-based nucleotides usually bind to the same site with similar affinity so kinases are not specific for ATP, but they may well not bind GTP or any other cellular components, so they may often be specific for adenine-based nucleotides. Thirdly, there are perhaps one thousand different proteins in humans that bind to ATP, and consequently ATP is not a specific ligand for a particular protein. On the other hand, a key feature of drug development is to obtain ligands that not only have high affinity but also bind to the target protein with much greater affinity than any other protein i.e., it is usually required that drugs are specific in this third sense.

Attaching a number to the specificity of a protein or ligand is difficult, or at least time consuming, as it requires comparative analysis. It is necessary to screening many ligands against a protein

or many proteins against a ligand. Often a ligand will bind to several evolutionarily or functionally related proteins and the practical investigation of specificity becomes a one of establishing how the ligand partitions between the desired target and each one of those related proteins. A level of specificity of 100-fold greater binding to the target than any other protein, where the proteins are at equal total concentration, would be considered "highly specific."

It is thought that some features of ligands and proteins confer specificity. A distinct arrangement of localized and directional interactions such as hydrogen bond donors/acceptors (discussed above) or charged groups are less likely to be complemented by chance than an array of apolar groups. Shape complementarity can also be a key factor in achieving specificity, with steric clashes created by single chemical groups of otherwise complementary molecules strongly discriminating against binding (30).

### 1.11 Nonspecific Binding

"Nonspecific" binding is used more narrowly to describe any binding of a ligand to sites other than that (or those) of interest; it might be more accurately called off-target binding. Typically, nonspecific/off-target binding is a consequence of the ligand having low affinity for a large number of binding-sites that exist in the experimental system. This results in binding curves having two components; a signal due to the ligand binding to the specific site and a signal due to binding to the nonspecific/low affinity sites. The first component has normal hyperbolic binding (*see* Subheading 2.1) reaching saturation at several-fold excess of ligand over protein and the second component shows a very slowly changing signal that saturates at much higher excess (100-fold) or may not appear to saturate at all within the range of concentrations studied. Nonspecific binding often gives false positive results in assays that are not subject to sophisticated analysis (many weak interactions can still bind a substantial amount of ligand) and may create errors or difficulties with analysis of titration data. However, nonlinear regression fits of binding curves (*see* Subheading 3) to a two component model can usually separate the components (31) if the off-target binding is substantial. Methods that measure an effect that is unique to a particular binding site (e.g., NMR chemical shifts of signals arising from an amino acid in the binding site) or competition experiments (i.e., in which a reference ligand known to bind exclusively at the site of interest is displaced by a test ligand) are relatively immune to nonspecific effects.

### 1.12 Chemical Modification and Fragment Approaches

From a basic research and biophysical perspective, the most valuable data is often obtained by studies of series of related reactions (e.g., changes that arise in the thermodynamics or kinetics of

binding a particular ligand due to mutation of residues in the protein or through binding many related ligands to a particular protein). Focusing on differences arising from small changes gives a greater chance of finding a plausible structural explanation for binding affinity or specificity for a particular ligand. At a superficial level it has become popular to perform alanine scanning mutagenesis on a protein (changing all its surface amino-acids one by one to alanine) and observing the effect on affinity of each mutation. It is found that residues in the binding site have most effect (and this is way of identifying the binding site in the absence of a structure) and that among those a small number of amino-acids will have the greatest effect—these are designated "hot-spot" residues (32). A similar approach can be taken from the ligand perspective. Either making modifications to series of ligands or fragmenting a ligand into each of its chemical substructures and binding each to the protein enables some assessment of the contribution of each part of the ligand to binding (33). However, it should be remembered that any change to the binding partners may be accompanied by changes in conformation and solvation of both the free and complexed molecules. Consequently, it cannot simply be assumed that large changes in affinity as a result of chemical changes definitively identify those groups that are the most important contributors in the native state; further structural characterisation is required as corroboration.

Fragments (molecules of <200 Da molecular weight representing common substructures of natural substrates and drugs) have also become a key tool in characterization of a protein's binding capabilities and in the development of novel ligands and drugs (34, 35). Distinct fragments that bind at neighboring sites on the surface of a protein may be chemically linked to create a high-affinity ligand or a fragment may be incrementally grown to include chemical groups providing additional specificity or affinity.

### 1.13 Data Resources for Protein–Ligand Interactions

There has been a strong trend to make the results of government-funded research available as widely as possible both to scientists and the public. Also web-resources have been created recently which replace some of the functions of commercial products, which used to be the main source of chemical information, with the aim of lowering the total cost to public sector scientists of obtaining information to further their research. Given an interest in a particular protein or ligand there is a good chance that something is already known about its interactions and much laboratory time can potentially be save by some desk-bound research. Table 1 describes some of the principal sources of information that are often used in studies of protein–ligand interactions.

**Table 1**
**Major sources of information on protein–ligand interactions**

| Resource | Contents | Utility | Reference |
|---|---|---|---|
| Blast | Sequences of all know proteins/genes | Search for proteins of similar sequence. Similarity to a region of a protein with known function may help infer ligand binding possibilities | blast.ncbi.nlm.nih.gov (36) |
| Pfam | Sequence motifs corresponding to functional and/or structural domains of proteins | Tools are available to identify domains in a user supplied protein sequence. Such domains typically have a conserved or related binding function | pfam.sanger.ac.uk (37) |
| CSD | 3D structures of >500,000 chemical compounds + tools and analyses based on these structures | Obtaining parameters to describe small molecules;<br>Statistical analyses of likelihood and geometry of interactions between chemical groups (Isostar);<br>Database of protein–ligand complexes derived from the PDB organized from a small molecule perspective to enable comparison of ligands and their interactions (Relibase) | www.ccdc.cam.ac.uk (38)<br>isostar.ccdc.cam.ac.uk (39)<br>relibase.ccdc.cam.ac.uk (40) |
| PDB | More than 80,000 protein structures, including >10,000 structures of protein-small molecule complexes | Searchable by protein sequence, ligand name and chemical substructures. Essential for modeling of ligand interactions, ligand docking, pharmaphocore development and attempts to understand results of mutation of protein or variation of ligand | www.pdb.org (41) |
| PDBsum | Graphical summaries of PDB entries | Maps of ligand interactions, structure descriptions and assessment of structure quality | www.ebi.ac.uk/pdbsum/ (42) |
| BindingDB | Binding data for >800,000 protein–ligand binding experiments (published and unpublished) | Comparative analysis for identical or homologous proteins or ligands, links to PDB structures where available | www.bindingdb.org (12) |

(continued)

**Table 1 (continued)**

| Resource | Contents | Utility | Reference |
|---|---|---|---|
| ChEMBL | Binding properties and bioactivities of >1,000,000 drug-like small molecules | Properties of compound or target of experimental interest; Identification of related ligands by structure similarity searching, comparative analysis of binding or pharmacokinetic behavior | www.ebi.ac.uk/chembl (43) |
| Brenda | Curated database of properties of enzymes organized by reaction type | Substrate specificity, affinity, inhibition constants, temperature and pH dependence, literature sources for previous work on the enzyme | www.brenda-enzymes.info/ (44) |
| Pubchem | Nonredundant database of available or previously synthesized compounds | Search for structurally related molecules, provides bioactivity and chemical data | pubchem.ncbi.nlm.nih.gov (45) |
| Zinc | Database of commercially available compounds including computer generated 3D structures | Search for structurally related molecules, create datasets for virtual screening | zinc.docking.org (46) |

## 2 Analysis of Interactions

This Subheading considers the definitions and concepts which are common to most of the experimental procedures that are discussed in later chapters. The discussion is specific to the simplest case of one-to-one binding, but all of the issues raised are applicable to more complicated cases (multiple binding sites, allostery, etc.).

### *2.1 Equilibrium Properties*

In most experimental methods, we measure a signal, $S$, arising from the complex, which changes with the fraction of total protein $[P]_T$ bound to ligand (or the closely related quantity of total amount of complex).

$$S = S_0 + f_b S_{sat}, \tag{10}$$

where $S_0$ is the signal with no ligand, $S_{sat}$ the signal when protein is saturated with ligand and the fraction bound is

$$f_b = \frac{[PL]}{[P]_T}. \tag{11}$$

The total amount of amount of protein and ligand is simply the amount of free species plus bound species

$$[\mathrm{P}]_\mathrm{T} = [\mathrm{P}] + [\mathrm{PL}] \tag{12}$$

$$[\mathrm{L}]_\mathrm{T} = [\mathrm{L}] + [\mathrm{PL}] \tag{13}$$

Consequently, the fraction bound can also be written

$$f_\mathrm{b} = \frac{[\mathrm{P}]_\mathrm{T} - [\mathrm{P}]}{[\mathrm{P}]_\mathrm{T}} = \frac{[\mathrm{L}]_\mathrm{T} - [\mathrm{L}]}{[\mathrm{P}]_\mathrm{T}}, \tag{14}$$

which means that the concentrations of free species can be written in terms of fraction bound and total protein and ligand concentrations

$$[\mathrm{P}] = [\mathrm{P}]_\mathrm{T}(1 - f_\mathrm{b}) \tag{15}$$

$$[\mathrm{L}] = [\mathrm{L}]_\mathrm{T} - f_\mathrm{b}[\mathrm{P}]_\mathrm{T}. \tag{16}$$

Substituting these quantities into the definition for $K_\mathrm{d}$ (Eq. 4) gives

$$K_\mathrm{d} = \frac{[\mathrm{P}]_\mathrm{T}(1 - f_\mathrm{b})([\mathrm{L}]_\mathrm{T} - f_\mathrm{b}[\mathrm{P}]_\mathrm{T})}{[\mathrm{P}]_\mathrm{T} f_\mathrm{b}} = \frac{(1 - f_\mathrm{b})([\mathrm{L}]_\mathrm{T} - f_\mathrm{b}[\mathrm{P}]_\mathrm{T})}{f_\mathrm{b}} \tag{17}$$

And rearranging gives a quadratic formula for fraction bound at equilibrium in terms of $K_\mathrm{d}$ and the total protein and ligand concentrations

$$f_\mathrm{b}{}^2[P]_\mathrm{T} - f_\mathrm{b}(K_\mathrm{d} + [\mathrm{L}]_\mathrm{T} + [\mathrm{P}]_\mathrm{T}) + [\mathrm{L}]_\mathrm{T} = 0. \tag{18}$$

This quadratic equation has two solutions, but only one gives physically reasonable results (i.e., that $f_\mathrm{b} = 0$ when $L_\mathrm{T} = 0$):

$$f_\mathrm{b} = \frac{(K_\mathrm{d} + [\mathrm{L}]_\mathrm{T} + [\mathrm{P}]_\mathrm{T}) - \sqrt{(K_\mathrm{d} + [\mathrm{L}]_\mathrm{T} + [\mathrm{P}]_\mathrm{T})^2 - 4[\mathrm{L}]_\mathrm{T}[\mathrm{P}]_\mathrm{T}}}{2[\mathrm{P}]_\mathrm{T}}. \tag{19}$$

Equation 19 (often called the tight-binding equation) describes the changes in fraction of protein bound to ligand for an interaction of given $K_\mathrm{d}$ as total ligand and/or protein concentration are changed. Equations 10 and 19 form the basis of all analysis of single-site protein–ligand titrations where they give a hyperbolic curve which tends to $S_\mathrm{sat}$ as the concentration of one component becomes relatively large (*see* Fig. 3).

### 2.2 $EC_{50}$

The total concentration of ligand at which half of the protein is bound is called the $EC_{50}$ (50 % effect concentration) of the reaction. The $EC_{50}$ is widely used to report results of assays since estimating the midpoint of a titration by inspection is straightforward. At the midpoint of a titration when half of the protein is bound to ligand $[\mathrm{P}] = [\mathrm{PL}]$ and $f_\mathrm{b} = ½$, substituting into Eq. 4 gives

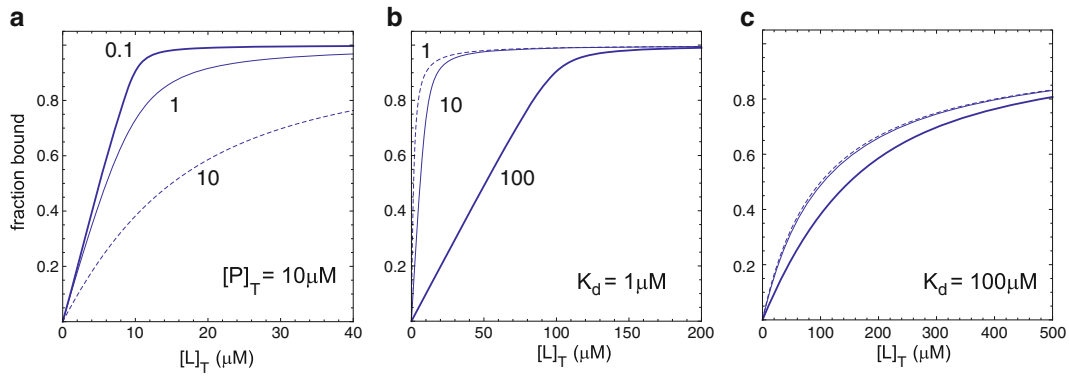


**Fig. 3** Properties of binding curves. The fraction of total protein bound to ligand (determined from Eq. 19) is shown as a function of the total ligand concentration, $[L]_T$, in a variety of circumstances. (**a**) For a fixed protein concentration ($[P]_T = 10$ μM) and $K_d = 0.1$, 1 and 10 μM. At the highest affinity, almost all added ligand binds to protein and the point of inflection (saturation) of the curve at ~10 μM ligand indicates the stoichiometry. Only when $K_d$ is similar to the total protein concentration is it likely to be accurately measurable using Eq. 21. (**b**) For a fixed and moderately high affinity, the fraction bound changes almost linearly with ligand/protein ratio for total ligand concentrations ($[P]_T = 1, 10, 100$ μM) above the $K_d$. (**c**) For a relatively low affinity interaction, fraction bound is almost independent of protein concentration for total protein concentration below the $K_d$ and does not saturate even at very high ligand to protein ratio ($[P]_T = 1, 10, 100$ μM as in **b**)

$$K_d = \frac{[P][L]}{[PL]} = [L]_{mid}. \tag{20}$$

The concentration of free ligand at which half the protein is bound is equal to the $K_d$. Consequently the $EC_{50}$ of a titration is often thought of as a reasonable estimate of $K_d$, however this is only the case if the free ligand and total ligand concentrations are similar. An alternative equation is obtained by substitution into Eq. 17. which gives

$$K_d = \frac{(1 - f_b)([L]_T - f_b[P]_T)}{f_b} = [L]_{T,mid} - \frac{[P]_T}{2} = EC_{50} - \frac{[P]_T}{2}. \tag{21}$$

The marginally simpler Eq. 20 is usually given in introductory textbooks as it is historically important and its theoretical treatment is more easily generalized to multiple binding sites. However, in many, if not most, experimental setups [L], the free ligand concentration, is not measured nor can it be well estimated from total ligand concentration, and Eq. 21, written in terms of total ligand concentration is much more useful. It is easy to see from Eq. 21 that using the $EC_{50}$ as an estimate of $K_d$ would not be correct unless the protein concentration was very much less than the $K_d$ (*see* Fig. 3c). At high protein concentration, the value of the $EC_{50}$ of the binding reaction has no special significance and simply tends to one half of the protein concentration, with the dissociation constant having little influence unless the interaction is very weak. Of course, it is

straightforward to obtain $K_d$ from the $EC_{50}$ by subtraction of $[P]_T/2$, but this can be subject to considerable error, e.g., if $[P]_T$ is 200 μM and $K_d$ is 1 μM the midpoint should be 101 μM ligand—if due to experimental error the measured ligand concentration is 103 μM at the midpoint (only a 2 % error) then the inferred $K_d$ of 3 μM is a factor of three (200 % error) from the true value. Consequently, effective assay setups require a protein concentration near the $K_d$ value to maximize the ability to detect weak binding while giving a reliably measurable $K_d$. Furthermore, fitting $K_d$ for the full titration curve is always more accurate than estimates from the midpoint as more data points are used to generate the result.

### 2.3 Competitive Binding and the $IC_{50}$

In many circumstances it is valuable to assess the competition between two ligands for binding to a protein. If one of these is a "reference" ligand with a known binding site and if titrating in another "test" ligand diminishes binding of the "reference," then it means that the test ligand either binds at the same site as the "reference" or at an allosteric site which influences the reference's binding site. Use of such a competition experiment largely rules out interference from nonspecific effects. Furthermore, it is sometimes very useful to construct assays around modulation of binding of a known ligand, since that will have consistent properties (e.g., UV or fluorescence spectrum) in each test reaction, rather than having to understand the properties of each of the many test ligands. Consequently, competitive inhibition (a.k.a. displacement) assays are very common in a medium to high-throughput contexts. If the reference compound is the protein's biological substrate or an analog thereof, then inhibition of substrate binding by the test ligand may form the basis of a drug discovery program.

Consider a situation of competitive binding where an inhibitor, I, directly competes with the reference ligand for a single binding site, i.e., the reaction

$$\mathrm{PL} \rightleftarrows \mathrm{P} + \mathrm{I} + \mathrm{L} \rightleftarrows \mathrm{PI}. \tag{22}$$

At equilibrium, both reactions obey Eq. 4 and if the dissociation constant for the inhibitor is $K_I$ then

$$[\mathrm{P}] = \frac{K_d[\mathrm{PL}]}{[\mathrm{L}]} = \frac{K_I[\mathrm{PI}]}{[\mathrm{I}]}, \tag{23}$$

and, thus

$$\frac{[\mathrm{PL}]}{[\mathrm{PI}]} = \frac{K_I[\mathrm{L}]}{K_d[\mathrm{I}]}, \tag{24}$$

Equation 24 shows that the effect of the inhibitor is simply proportional, the greater the affinity or concentration of free inhibitor [I], the lower the ratio of ligand is bound.

However, since [I] will itself depend upon the $K_I$ and $K_d$ and the relative concentrations of I and L, the equilibrium situation in terms of the total concentrations of the reactants is more complicated than the single ligand case in Subheading 2.1. Only relatively recently has a full analysis been made of the cubic equations, the work of Wang (47) means that it is now possible to carry out nonlinear fitting using computer software for competitive binding. However, such analysis is as yet rather infrequently carried out in practice.

Probably as a result of tradition and its superficial simplicity, it is most common to characterize competitive inhibition in terms of an $IC_{50}$ value. Reflecting the definition of $EC_{50}$, the $IC_{50}$ of an inhibitor is the concentration of inhibitor at which the maximal effect produced by reference ligand binding is reduced by 50 %. This makes for a simple assay format in which a signal from the ligand complex is monitored and inhibitor titrated until the signal is reduced by half and this practical simplicity explains the wide use of $IC_{50}$ in studies of inhibitor series. However, although the relative values of $IC_{50}$ for a particular assay setup do provide a useful ranking of the relative affinity of a series of ligands, it is generally difficult or impossible to convert $IC_{50}$ values to $K_I$ or $\Delta G_{bind}$ values. Commonly used expressions for making such conversions (48) are subject to very restrictive conditions under which they are valid, and although alternatives may be applicable in wider circumstances (49) such conversion is generally not recommended for quantitative study of protein–ligand binding. Having identified "interesting" inhibitors in a high-throughput assay, experimenters should separately measure their binding reactions in a noncompetitive experiment or investigate and analyze the concentration dependence in a competition experiment more fully (47, 50) to extract the inhibitor's intrinsic binding parameters. Further details on the theoretical analysis and practical aspects of competition experiments are given in Chapter 3.

### 2.4 Approach to Equilibrium

Consider a situation in which an amount of ligand is added to protein. Initially, every protein molecule will be unbound ($[P] = [P]_T$) and encounters between ligand and protein frequently lead to binding. As complexes are formed then there are fewer free ligands, and those ligands will less frequently encounter free protein, consequently the rate of complex formation will slow. Simultaneously, the number of complexes is increasing, and since each complex dissociates with a constant rate $k_{off}$, the total dissociation rate rises. The association rate continues falling and dissociation rate rising until equilibrium is reached when association and dissociation rates are equal. At any point in time, $t$, the rate of formation of the complex is the difference between association and dissociation rates

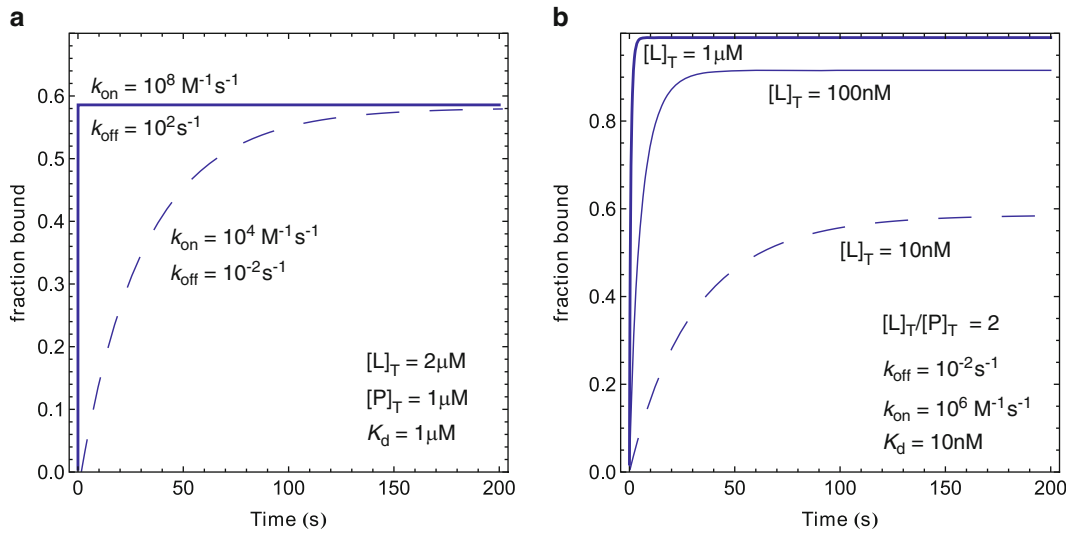


**Fig. 4** The approach to equilibrium. (**a**) For a typical $K_d = 1$ μM the time to achieve equilibrium changes almost in proportion to the on rate constant from ~0.01 s to ~100 s as it is varied from a typical $10^8$ $M^{-1}$ $s^{-1}$ to a slow $10^4$ $M^{-1}$ $s^{-1}$, such as might be encountered if binding required a significant conformational change of the protein. (**b**) For a typical high-affinity ligand, time to equilibrium slows dramatically at concentrations near (or are below) $K_d$

$$\begin{aligned}\frac{d[PL]}{dt} &= k_{on}[P][L] - k_{off}[PL] = k_{on}([P]_T - [PL])([L]_T - [PL]) - k_{off}[PL] \\ &= k_{on}[P]_T[L]_T - (k_{on}([P]_T + [L]_T) + k_{off})[PL] + k_{on}[PL]^2.\end{aligned} \tag{25}$$

Shortly after ligand is injected little complex will have formed, thus the term in $[PL]^2$ can be ignored, and the rate of formation of complex approximately falls in direct proportion to amount of complex present. Consequently, at least initially or in any circumstance where [PL] is small, the approach to equilibrium is described by a simple exponential, i.e.,

$$f_b(t) \propto 1 - e^{-(k_{on}([P]_T+[L]_T)+k_{off})t}. \tag{26}$$

The time constant for the approach to equilibrium, $\tau_{relax} = 1/(k_{on}([P]_T + [L]_T) + k_{off})$, turns out to be a good approximation to the true value in a wide variety of circumstances (*see* **Note 4**). The fraction of protein bound will usually achieve more than 98 % of its equilibrium value in 4 $\tau_{relax}$ and in practice this time often depends largely upon $k_{on}$ and the total concentration of protein and ligand since $k_{off}$ is relatively small.

Experiment design for equilibrium measurements should take into account the time taken to reach equilibrium. For a typical protein–ligand binding assay with 1 μM protein, 2 μM ligand and a $K_d$ of 1 μM the time to equilibrium can vary dramatically with the kinetic properties of the protein (*see* Fig. 4). Whereas for a typical $k_{on}$

of $10^8$ $M^{-1}$ $s^{-1}$ and a $k_{off}$ of 100 $s^{-1}$, equilibrium is achieved in approximately 0.01 s and is unlikely to be of practical importance, for a system with the same $K_d$ but with slow kinetics equilibrium may not be achieved until more than 100 s have elapsed (*see* Fig. 4). Working at very low protein concentration also extends the time to equilibrium. In the case of a competition experiment where, at the start of the experiment, the reference ligand is already bound to the protein, then the initial on-rate of the test ligand is diminished in proportion to the degree of saturation and also may substantially depend on the off-rate of the reference compound. Because off-rates are usually slow, competition experiments are particularly prone to inaccurate results due to failure to achieve equilibrium. Consequently, when establishing an experiment design it is important to try several delays before measurements are made in order to detect any possible issues (*see* Chapter 3 for further discussion).

## 3 Dealing with Binding Data

### 3.1 Nonlinear Regression Analysis

All of the equations that describe binding reactions in terms of total protein and total ligand concentrations are inherently "nonlinear" in respect of their important parameters (i.e., doubling $K_d$ does not halve fraction bound) (**Note 5**). The standard computational methods for fitting nonlinear equations to data (called nonlinear regression) use iterative approaches which proceed from an initial guess of the parameter values and systematically modify them to minimize the deviation between experimental datapoints and the values given by the parameterized equation. Although not the only statistically justifiable approach, in an overwhelming majority of cases a least-squares criterion is used to evaluate the minimum deviation between the experimental values ($d_{i,expt}$) and those calculated with the current set of parameters ($d_{i,calc}$), i.e.,

$$\mathrm{Min}\left[\sum_i \frac{\left(d_{i,expt} - d_{i,calc}\right)^2}{\sigma_i^2}\right], \tag{27}$$

is sought, where $\sigma_i$ is the experimental standard deviation of the $i$th datapoint (**Note 6**).

Computer programs for iterative refinement of the parameter values most commonly implement the Levenberg-Marquardt (LM) algorithm (51). This algorithm relies on being able to calculate how the least-squares deviation values change with small variations in the current parameters (i.e., the partial derivatives of the equation with respect to the parameters) and making changes in the parameters in the direction determined to lead to most rapid reduction in the deviation. The LM method is an adaptive combination of a straightforward gradient minimization and the Gauss-Newton (GN) approach (which is also sometimes encountered) and is

designed to be more efficient than either (51, 52). It is important to realize that no nonlinear regression method is guaranteed to find the parameters giving the minimum deviation in all circumstances. Both LM and GN methods generally work well, but if one fails to find a good fit then the other may succeed since they will follow different routes in searching for the minimum.

All nonlinear regression methods rely on having a reasonably good initial guess for the parameters and on the equation being fitted actually being relevant to the real behavior of the system being studied. Any program may fail to find a genuinely good fit if either of these conditions are not met. Usually, it is the user of the program who needs to spot any problems. It is vitally important not to blindly believe the best-fit parameters. A program will (almost) always give an answer for "best-fit" parameters, but "best" in this sense means "best that the program could do in the circumstances"! It is always necessary to inspect the curve described by the best fit parameters, to see if it passes through the data points. The residuals (i.e., the values of $d_{i,expt} - d_{i,calc}$) should be inspected to see if they vary in any systematic way with concentration during a titration. If they do, then sources of systematic error in the experiment (Subheading 3.2) and alternative binding models should be considered. One caveat however, is that more complex binding models almost always give better fits to data with smaller residuals; consequently improved residuals alone do not make the more complicated model better. Statistical testing is required to establish which is the most likely model in those cases where the simplest model is unsatisfactory. Such decisions are considered in detail in the excellent introductions by Tellinghuisen (53) and Johnson (54).

### 3.2 Sources of Error in Estimated Binding Parameters

Experiments and their analysis can be afflicted by several sources of systematic and/or sampling errors. Systematic errors can include miscalibration of instruments, failure to observed protocols or instrument instructions precisely, loss of protein activity (e.g., due to progressive aggregation), inaccurate affinities for the reference compound in competition experiments, variation in solution conditions between batches of protein and failure to achieve equilibrium. Sampling errors arise from variations in instrument performance from moment to moment due to electrical noise or mechanical variation (e.g., in pipetting).

Systematic errors can often be identified by the availability of one or more control experiments. In fact, one should not consider carrying out biophysical analysis without first establishing a positive control experiment for the instrument (e.g., using some inexpensive commercially available protein and a known ligand). Protocols should be practised with the control until you can achieve consistent results. Any instrument variation over time can also be checked with this control experiment. Negative controls should also be performed for the "target" protein, i.e., carry out the binding protocol

without ligand, this should show up any effect from protein aggregation or interaction of protein with buffer components. If a positive control is available for the target protein, i.e., a known ligand is available in sufficient quantity, then that is also valuable as it can help identify any batch-to-batch variation in the protein preparation.

In terms of absolute measurements of binding constants, the most common sources of error are concentration measurements of the protein and/or ligand, e.g., *see* ref. 55. Protein concentration measurement is discussed in Chapter 2. The critical point is that most concentration measurement methods have some level (perhaps 5–10 %) of error in the measurement of absolute protein concentration due to their measured signal being dependent on the protein sequence in a not entirely predictable manner. In critical cases, a sample of the stock protein solution should be sent for commercial quantitative amino acid analysis and any routine laboratory method benchmarked against the results of that analysis. Ligands are most commonly received from synthesis as dry powders. Because it is difficult to accurately and consistently weigh small quantities of ligand for individual experiments, stock solutions sufficient for a whole series of experiments should be made wherever possible. For commercially purchased ligands, it may be worthwhile having compounds supplied as solutions of certified composition.

Even if concentrations of protein and ligands are known accurately, it is not necessarily the case that all protein or all ligand in the sample are competent to bind; there could be impurities in the ligand or a portion of the protein could be misfolded. This can be difficult to detect as small (<20 %) losses of activity may not be easily distinguished. Quantitation of the active proportion of the reagent is best supported by some orthogonal technique, e.g., ligand purity can be checked by HPLC or mass spectrometry, and NMR spectra of a protein with and without a control ligand can eliminate the possibility of misfolding. In general, the best guard against errors from loss of activity in binding is a thorough physical characterization of the protein. If a positive control ligand-binding reaction is available for the target protein, then the stoichiometry of that reaction (e.g., as determined in ITC) may enable percent activity to be measured and other results can then be suitably corrected. Furthermore, it is often the case that the absolute values of $K_d$, etc., are less important than the relative values in a series of ligands being examined. For example, the relative affinity or the ranking of affinities is important for decision making in drug development. In such cases, when working with a protein stock that has a consistent active concentration, the relative systematic errors in parameters will often be smaller than the absolute systematic error.

Data analysis becomes more susceptible to measurement error at extremes of high- and low-affinity binding (as the binding hyperbola

becomes very rectangular at high affinity, small measurement errors can lead to large changes in best-fit parameters; for low affinity, the binding curve becomes very shallow and it is difficult to saturate the protein or distinguish any nonspecific binding). In the high-affinity case, working at lower concentration of reactants may improve the quality of fit, but at the expense of signal to noise. Equally, low affinity interactions are more readily quantitated at higher concentration. There is an optimal compromise for each protein–ligand system and experimental method and this may be worth identifying for large scale assays and difficult cases.

### 3.3 Analysis of Random Errors

Random sampling errors in individual measurements naturally give rise to deviations of the best-fit parameters from their true values. It is important to establish the likely size of those deviations if the best-fit parameters are to be used in decision making or in validating computational models of individual protein–ligand interactions or in constructing models of systems of interactions. Ideally, titration experiments should be repeated several times and the values of the best-fit parameters determined for each titration averaged and the variation of each value used to estimate the error in mean value of each parameter following standard statistical methods (53). Certainly, publication quality studies will often require duplicates or triplicates.

Restrains on time in a high-throughput context, or availability of reagents more generally, may mean that no or only a few experiments are repeated for a given set of experimental conditions. In this circumstance some estimate of likely error in the derived parameters is still both possible and necessary. Such estimates can be obtained even from a single titration dataset by virtue of the fact that each measurement likely contains some random error, and thus the data points are unlikely to lie precisely on the true binding curve, but will be scattered about it. Thus, the scatter of the data points around the curve described by the best-fit parameters, which is our best-estimate of the true curve, will provide a measure of error in the data points and, via further analysis, the errors in the parameters.

For nonlinear equations, there is no exact theory that relates errors in the data to errors in the parameters. Most, if not all, commercial data analysis packages give a linear approximation called the "asymptotic standard error" as a measure of parameter error. As a rule of thumb, only where the asymptotic error is small (say $<10$ % of the parameter value) is it likely to be a reasonable estimate of the overall error in the parameters (53). Preferably, and certainly where sampling errors are large, a more robust analysis via Monte Carlo (MC) resampling or bootstrapping should be performed (**Note 7**). The principle of Monte Carlo error analysis is simple; artificial data sets are created by adding random errors to each experimental data point and then the best fit equation to each

artificial data set is found. The errors in individual data points may be estimated from their residuals, which will naturally underestimate the error in some data points and overestimate it in others, but if there are sufficient datapoints this will "average out" (a related alternative approach where errors on datapoints are expected to be similar is to repeatedly randomly swap some of the residuals—an approach known as bootstrapping (52)). Each regression analysis of the artificial datasets will give slightly different parameters to the original. Repeat this process many times and you will find many values for parameters that best fit these artificial data. Their mean values will be the same as the fit to the original experimental data, and their variance/standard deviation will be a good estimate of the errors in the parameters that arise from errors in the data points. However, it should still be borne in mind that the errors estimated in this way will only reflect the measurement stage of the protocol, and replicates of the entire procedure may highlight variations that occur elsewhere.

### 3.4 Conclusion

The three most valuable in attributes for the experimental analysis of protein-ligand complexes are caution, curiosity and experience. The methods themselves are usually straight forward. Instruments come with manuals, and other chapters in this book provide further details of protocols and experimental strategies that are important to achieving a successful outcome. Data analysis routines are implemented in a variety of software and are generally straightforward to use. Although, the output of software must be treated with caution and some understanding of its meaning in order to enable realistic reporting of results. In practice, most difficulties arise from features of the protein's behavior, such as instability or multiple binding sites, sensitivity to pH, salt or other buffer components. Experience of "normal" experiments is essential to enable the experimenter to identify the "unexpected" results and formulate hypotheses as to their origin. In this regard, replicates and control experiments are invaluable for accurate quantitative work and generating results that future users of biophysical data can rely on.

## 4 Notes

1. Whereas Eq. 6 is almost universally cited, because taking the logarithm of $K_d$ (which has units of Molar) is not strictly allowed, some authors modify Eq. 6 to include an additional term within the logarithm so that $\Delta G_{bind}$ is defined at a hypothetical 1M standard state. In practice, provided Molar units of concentration are used throughout, this difference has no impact on the numerical values.

2. Most protein–ligand experiments are conducted under standard conditions for chemistry, i.e., 25 °C (298.15K) and 1 atmosphere pressure. Under these conditions $RT \approx 2.4789569$ kJ mol$^{-1}$. For conversion of joules to calories, which are still widely used in the USA, divide by 4.184. It is advisable not round any numbers until the very final step of any calculation—rounding errors account for inaccuracies in more than 10 % of publications (16).
3. Formally, the enthalpy change, ΔH, includes both the change in internal energy or heat (ΔE) and the work done against external pressure by volume changes resulting from the reaction. In biomolecular reactions, the external work is usually very small compared to ΔE and, as here, is usually ignored.
4. The general solution for the approach to equilibrium described by Eq. 25 is

$$f_b(t) = \frac{1}{2[\mathrm{P}]_\mathrm{T}}\left(K_\mathrm{d} + [\mathrm{L}]_\mathrm{T} + [\mathrm{P}]_\mathrm{T} - \mathrm{Re}\left(\gamma \tan\left(\arccos\left(\frac{1}{2\sqrt{[\mathrm{L}]_\mathrm{T}[\mathrm{P}]_\mathrm{T}}}\gamma\right) - \frac{k_\mathrm{on} t}{2}\gamma\right)\right)\right), \tag{28}$$

where $\gamma = \sqrt{4[\mathrm{L}]_\mathrm{T}[\mathrm{P}]_\mathrm{T} - \left(K_\mathrm{d} + [\mathrm{L}]_\mathrm{T} + [\mathrm{P}]_\mathrm{T}\right)^2}$ and Re (..) signifies the real part of the expression. Equation 28 converges to the equilibrium expression for $f_b$ (Eq. 19) at large values of the time, $t$, however, it is rather complicated to compute. If we are only interested in the initial rate or situations when concentration of complex formed is low compared to the total reactants, then an approximate solution is

$$f_b(t) = \frac{k_\mathrm{on}[\mathrm{L}]_\mathrm{T}}{k_\mathrm{on}\left([\mathrm{P}]_\mathrm{T} + [\mathrm{L}]_\mathrm{T}\right) + k_\mathrm{off}}\left[1 - \mathrm{e}^{-\left(k_\mathrm{on}\left([\mathrm{P}]_\mathrm{T}+[\mathrm{L}]_\mathrm{T}\right)+k_\mathrm{off}\right)t}\right]. \tag{29}$$

This does not converge to the correct equilibrium value for $f_b$ except in the weak binding limit, however, the rate of approach is rather similar to the true value across a wide range of conditions. Another approximation is frequently encountered in the enzyme literature (56), i.e., that the protein concentration is negligibly small and thus

$$f_b(t) = \frac{k_\mathrm{on}[\mathrm{L}]_\mathrm{T}}{k_\mathrm{on}[\mathrm{L}]_\mathrm{T} + k_\mathrm{off}}\left(1 - \mathrm{e}^{-\left(k_\mathrm{on}[\mathrm{L}]_\mathrm{T}+k_\mathrm{off}\right)t}\right). \tag{30}$$

However, in general, Eq. 30 is a much less accurate reflection of the true rate than Eq. 29, and, since it is also not much simpler, its use is discouraged.
5. Whereas it is essential to utilize the general (nonlinear) binding or kinetic equations in most quantitative studies of protein–ligand interactions, in some cases, notably when studying enzymes,

experimental circumstances prevail such that some simplifications can be made to analysis. Many enzyme reactions are carried out in the presence of a large excess of ligand and/or where the concentration of protein is very low ($<0.01 \times K_d$), and consequently the equations describing reactions can be simplified to assume that free ligand concentration is very little change by formation of the complex. This is the scenario in which much of the classical analysis of enzymes was developed in the period 1910–1940, and these methods of analysis still prevail in certain application areas. These linearised approaches, which provide for simpler graphical or linear regression analysis, are described in detail in Copeland's excellent textbooks (56, 57). Be careful to ensure that the conditions for their validity are met before using them (52, 58).

6. In the $\chi^2$ measure (Eq. 27) used to identify the parameters giving the best fit to the data, each term in the sum is weighed by the experimental standard deviation in each data point ($\sigma_i$). In principle, this means that values which are experimentally more precisely known are more important in the fitting process, which is a good idea. In practice, however, the $\sigma_i$ may not be accurately known (e.g., the error in the estimate of the standard deviation is 50 % for three repeats). Consequently, in cases where the $\sigma_i$ values are unknown or vary by less than a factor of 2–3, it is more reasonable to use a uniform weight of $\sigma = 1$ in identifying the best-fit. Conversely, if there is some *a priori* knowledge that there will be a systematic variation in the deviation (e.g., that $\sigma_i$ depends on [PL]) then appropriate relative weights can be used without determining the actual standard deviations.
7. Monte Carlo analysis is accessible via the scripting capabilities of several commercial data analysis packages (GraphPad Prism, GraphPad Software; Kaliedagraph, Synergy Software; Origin 8, OriginLab), and can be readily implemented in general mathematical packages (Mathematica, Wolfram Research; Matlab, TheMathWorks) or even in a spreadsheet program (59).

## References


1. Andrews SS, Bray D (2004) Stochastic simulation of chemical reactions with spatial resolution and single molecule detail. Phys Biol 1:137–151
2. Resat H, Petzold L, Pettigrew MF (2009) Kinetic modeling of biological systems. Methods Mol Biol 541:311–335
3. Davis ME, Madura JD, Sines J, Luty BA, Allison SA, McCammon JA (1991) Diffusion-controlled enzymatic reactions. Methods Enzymol 202:473–497
4. Northrup SH, Erickson HP (1992) Kinetics of protein–protein association explained by Brownian dynamics computer simulation. Proc Natl Acad Sci U S A 89:3338–3342
5. Camacho CJ, Weng Z, Vajda S, DeLisi C (1999) Free energy landscapes of encounter complexes in protein–protein association. Biophys J 76:1166–1178
6. Schreiber G, Fersht AR (1996) Rapid, electrostatically assisted association of proteins. Nat Struct Biol 3:427–431

7. Held M, Noé F (2011) Calculating kinetics and pathways of protein–ligand association. Eur J Cell Biol 91:357–364
8. Schreiber G, Haran G, Zhou HX (2009) Fundamental aspects of protein–protein association kinetics. Chem Rev 109:839–860
9. Kiel C, Selzer T, Shaul Y, Schreiber G, Herrmann C (2004) Electrostatically optimized Ras-binding Ral guanine dissociation stimulator mutants increase the rate of association by stabilizing the encounter complex. Proc Natl Acad Sci U S A 101:9223–9228
10. Gaboulline RR, Wade RC (2002) Biomolecular diffusional association. Curr Opin Struct Biol 12:204–213
11. Gutfreund H (1995) Kinetics for the life sciences: receptors, transmitters and catalysts. Cambridge University Press, Cambridge, UK
12. Liu T, Lin Y, Wen X, Jorrisen RN, Gilson MK (2007) BindingDB: a web-accessible database of experimentally determined protein–ligand binding affinities. Nucleic Acids Res 35: D198–D201
13. Ladbury JE, Klebe G, Freire E (2010) Adding calorimetric data to decision making in lead discovery. Nat Rev Drug Discov 9:23–27
14. Olsson TSG, Ladbury JE, Pitt WR, Williams MA (2011) The extent of enthalpy-entropy compensation in protein–ligand interactions. Protein Sci 20:1607–1618
15. Dunitz JD (1995) Win some, lose some: enthalpy-entropy compensation. Chem Biol 2:709–712
16. Olsson TSG, Williams MA, Pitt WR, Ladbury JE (2008) The thermodynamics of protein–ligand interaction and solvation: Insights for ligand design. J Mol Biol 384:1002–1017
17. De Godoy LM et al (2008) Comprehensive mass-spectrometry-based proteome quantification of haploid versus diploid yeast. Nature 455:1251–1254, data in Supp. Fig. 5 with haploid cell volume of 42 $\mu m^3$
18. Pawson T, Nash P (2003) Assembly of cell regulatory systems through protein interaction domains. Science 300:445–452
19. Kavanagh KL, Jörnvall H, Persson B, Oppermann U (2008) Medium- and short-chain dehydrogenase/reductase gene and protein families: the SDR superfamily: functional and structural diversity within a family of metabolic and regulatory enzymes. Cell Mol Life Sci 65:3895–3906
20. Kahraman A, Richard J, Morris RJ, Laskowski RA, Favia AD, Thornton JM (2010) On the diversity of physicochemical environments experienced by identical ligands in binding pockets of unrelated proteins. Proteins 78:1120–1136
21. Williams MA, Ladbury JE (2003) Hydrogen bonds in protein–ligand complexes. In: Boehm H-J, Schneider G (eds) Molecular recognition in protein–ligand interactions. Wiley, New York
22. McDonald IK, Thornton JM (1994) Satisfying hydrogen bond potentials in proteins. J Mol Biol 238:777–793
23. Williams MA, Goodfellow JM, Thornton JM (1994) Buried waters and internal cavities in monomeric proteins. Protein Sci 3:1224–1235
24. Fersht AR et al (1985) Hydrogen bonding and biological specificity analysed by protein engineering. Nature 314:235–238
25. Hann MM, Leach AR, Harper G (2001) Molecular complexity and its impact on the probability of finding leads for drug discovery. J Chem Inf Comput Sci 41:856–864
26. Ferrara P, Gohlke H, Price DJ, Klebe G, Brooks CL 3rd (2004) Assessing scoring functions for protein ligand interactions. J Med Chem 47:3032–3047
27. Sotriffer CA, Sanschagrin P, Matter H, Klebe G (2008) SFCscore: scoring functions for affinity prediction of protein–ligand complexes. Proteins 73:395–419
28. Jorgensen WL (2009) Efficient lead discovery and optimization. Acc Chem Res 42:724–733
29. Ladbury JE, Williams MA (2004) The extended interface: measuring non-local effects in biomolecular interactions. Curr Opin Struct Biol 14:562–569
30. Koh JT (2002) Engineering selectivity and discrimination into ligand-receptor interfaces. Chem Biol 9:17–23
31. Mendel CM, Mendel DB (1985) "Non specific" binding a problem and a solution. Biochem J 228:269–272
32. Zerbe BS, Hall DR, Vajda S, Wjitty A, Kozakov D (2012) Relationship between hot spot residues and ligand binding hot spots in protein–-protein interfaces. J Chem Inf Model. doi:10.1021/ci300175u
33. Ciulli A et al (2006) Probing hot spots at protein–ligand binding sites: a fragment-based approach. J Med Chem 49:4992–5000
34. Ciulli A, Abell C (2007) Fragment-based approaches to enzyme inhibition. Curr Opin Biotechnol 18:489–496
35. Hadjuk PJ, Greer J (2007) A decade of fragment-based drug design: strategic advances and lessons learned. Nat Rev Drug Discov 6:211–219
36. Altschul SF, Gish W, Miller W, Myers EW, Lipman DJ (1990) Basic local alignment search tool. J Mol Biol 215:403–410

37. Punta M et al (2012) The Pfam protein families database. Nucleic Acids Res 40:D290–D301
38. Allen FH et al (2002) The Cambridge structural database: a quarter of a million crystal structures and rising. Acta Cryst B58:380–388
39. Bruno IJ et al (1997) Isostar: a library of information about non-bonded interactions. J Comput Aided Mol Des 11:525–537
40. Günther J, Bergner A, Hendlich M, Klebe G (2003) Utilising structural knowledge in drug design strategies: applications using Relibase. J Mol Biol 326:621–636
41. Berman HM et al (2000) The protein data bank. Nucleic Acids Res 28:235–242
42. Laskowski RA (2009) PDBsum new things. Nucleic Acids Res 37:D355–D359
43. Gaulton A et al (2011) ChEMBL: a large-scale bioactivity database for chemical biology and drug discovery. Nucleic Acids Res 40: D1100–D1107
44. Scheer M et al (2011) BRENDA, the enzyme information system in 2011. Nucleic Acids Res 39:670–676
45. Bolton E, Wang Y, Thiessen PA, Bryant SH (2008) PubChem: integrated platform of small molecules and biological activities. In: *Annual Reports in Computational Chemistry*, Vol 4, American Chemical Society, Washington
46. Irwin JJ, Shoichet BK (2005) ZINC—a free database of commercially available compounds for virtual screening. J Chem Inf Model 45:177–182
47. Wang Z-X (1995) An exact mathematical expression for describing competative binding of two different ligands to a protein molecule. FEBS Lett 360:111–114
48. Cheng Y-C, Prushoff WH (1973) Relationship between the inhibition constant ($K_I$) and the concentration of inhibitor which causes 50 % inhibition ($IC_{50}$) of an enzymatic reaction. Biochem Pharm 22:3099–3108
49. Lazareno S, Birdsall NJM (1993) Estimation of antagonist Kb from inhibition curves in functional experiments: alternatives to the Cheng-Prushoff equation. Trends Pharm Sci 14:237–239
50. Sigurskjold BW (2000) Exact analysis of competition ligand binding by displacement isothermal titration calorimetry. Anal Biochem 277:260–266
51. Press WH, Teukolsky SA, Vetterling WT, Flannery BP (2007) Numerical recipes: the art of scientific computing, 3rd edn. Cambridge University Press, New York
52. Johnson ML (2008) Non-linear least-squares fitting methods. Meth Cell Biol 84:781–805
53. Tellinghuisen J (2008) Stupid statistics! Meth Cell Biol 84:739–780
54. Straume M, Johnson ML (1992) Analysis of residuals: criteria for determining goodness of fit. Meth Enzymol 210:87–105
55. Myszka DG et al (2003) The ABRF-MIRG '02 Study: assembly state, thermodynamic, and kinetic analysis of an enzyme/inhibiter interaction. J Biomol Tech 14:247–269
56. Copeland RA (2005) Evaluation of enzyme inhibitors in drug discovery: a guide for medicinal chemists and pharmacologists. Wiley, New York
57. Copeland RA (2000) Enzymes: a practical introduction to structure, mechanism and data analysis. Wiley, New York
58. Johnson ML (1992) Why, when, and how biochemists should use least squares. Anal Biochem 206:215–225
59. Delahunty MD, Mack JPG (1993) A general method of curve-fitting and error analysis using a spreadsheet: determination of the binding constants of tight binding ligands in variable volume assays. Comput Appl Biosci 9:127–131